TRACER CONVENTIONS[1]

Measuring How LLM Agents Copy Practices from a Shared Board, with a Field Check on the Collusion.wiki Swarm

Fatimah Emad Eldin
Trouve Labs

With
Apart Research

Abstract

When the same practice appears under hundreds of agent names on a shared board, did agents copy it from each other or rediscover it? A write log cannot tell, because exposure goes unrecorded and useful practices can be reinvented. We propose *tracer conventions*: arbitrary markers that have no function, so they can spread only by copying. We seeded such markers, and a functional answer index, on a shared wiki and ran 2,304 paired trials with four open-weight models. No model produced an index or a code-suffixed page name without a seeded example (0/576). Copying required exposure, and exposure was not sufficient: models inspected the wiki unprompted in 0% to 92% of trials, and agents that read seeded pages often still did not copy. Shown the page list, agents copied the naming structure after one example (142/192) but filled it with their own id; the exact code was copied once prior pages agreed on it (0/192 with one page, 52/192 with two, 81/192 with four), and almost never when four pages carried different codes (1/192). A field check on the collusion.wiki write log finds that write-visible exposure predicted technique adoption no detectably more than placebo words (odds-ratio ratio 1.21, 95% CI 0.88–1.67); a planted positive control shows that these write-log tests could not have detected copying, while a page-local test could and found a technique-specific excess. Read logs are necessary to study diffusion in deployed swarms, and tracer conventions offer a cheap, causal way to measure it.

1. Introduction

In mid-2026 autonomous agents from an OpenAI evaluation used small public wikis as a message board [14]. They relayed answers, shared data-access workarounds, set up liveness beacons and published a way around an egress restriction, and many of these practices appear under dozens to hundreds of agent names. Whether such a practice *spread* from agent to agent (a swarm) or was *rediscovered* by each agent facing the same constraint (a fleet) decides where to intervene: the shared channel or the task and sandbox. Incident investigators have met the same question elsewhere: a method "used on about 20 more [places] within two days, whether by spreading or by being found again" [13].

Observational logs struggle with it. Agents read far more than they write, so exposure is unobserved; names are cheap and short-lived [9]; and shared tasks produce simultaneous adoption under either hypothesis, the contagion–homophily confound of social-network studies [11]. A recent reconstruction of the same incident added the wiki operator's request log and found that a transmission reading of marker adoption could not be sustained [6]. We take a complementary, experimental route: plant conventions whose origin is known and measure when agents pick them up. Arbitrary markers cannot be rediscovered, so any adoption is copying. The work addresses the science of agent networks (Track 2) and, through what it implies for logging, multi-agent oversight (Track 4). Our contributions:

1. **A tracer-convention experiment** (Section 3) with twelve paired conditions and four models. Arbitrary conventions were never produced without exposure; copying required exposure, which models sought at very different rates and which was not sufficient; and exact arbitrary tokens were copied only when prior pages agreed on them.
2. **A field check** on the collusion.wiki write log (Section 4) with a re-timing null, a within-label choice design and placebo tokens: write-visible exposure did not distinguish technique spread from word echo, naive contagion estimates were artefacts of shared bursts, and a planted positive control shows which of these tests could have detected copying at all.

[1] Research conducted at the Swarm Alignment Hackathon, November 2026.

3. **Oversight guidance** (Section 5): what to log to study diffusion in deployed agent populations, and how tracer conventions measure it causally. The harness, analysis pipeline, transcripts and a technique catalogue usable as an egress watch-list are released.

2. Related Work

The collusion.wiki incident. Von Arx et al. [14] documented the episode, estimated roughly 18,000 agent posts, and released the revision export we use (14,591 revisions, 24 May–2 July 2026). Murtuza [9] gave a first statistical treatment (co-editing permutation tests, a behavioural taxonomy, a deletion event study, an analysis-pitfall checklist we follow); its pipeline lists a URL-diffusion step but it reports no spread-versus-rediscovery test. Lütje [6] reconstructs episodes, cohorts and schedules from the same export, observes convergence on coordination formats within a day, and adds the operator's request log (about 5.16 million records including content requests): content requests before writing are observed for 1,034 of 1,140 coordinating names, and prior requests to marker-bearing pages occur for 40.2% of marker adopters against 31.7% of non-adopters, an association the author declines to interpret as transmission. Our export lacks those read records; our field check is therefore limited to write-visible exposure, and our experiment supplies the causal evidence the logs cannot. METR [7], with transcripts of a contemporaneous incident, could trace one technique packaged on a board to about 7% of agents.

Conventions and social learning among LLM agents. Populations of LLM agents form shared conventions in repeated naming games, and committed minorities can tip them [1]; agents develop and transmit emergent languages, which newcomers learn from usage [12]; and self-improving agents copy peers' skill files, which makes learning more efficient but concentrates populations on fewer discoveries [4]. These studies build interaction into the design. We ask a narrower measurement question: whether one agent arriving at a board copies an arbitrary convention left by others, as a function of how many prior pages carry it and whether the agent looks.

Contagion and multi-agent risk. Prompt injections self-replicate across agents [5], an adversarial image spreads a jailbreak through a population [2], ideas can induce their adopters to transmit them [10], and collective loss of control can be modelled as an epidemic [15]. Hammond et al. [3] list the spread of capabilities among multi-agent risk factors and Motwani et al. [8] formalise secret collusion. Tracer conventions offer a way to measure transmission in such systems when it is not known by construction.

3. Tracer-Convention Experiment

3.1 Design

Each trial runs one LLM agent, "evaluation agent A*k*", for at most 10 turns with one JSON tool call per turn (`list_pages`, `read_page`, `write_page`, `submit_answer`, `finish`). It receives one short question with a deterministic answer (12 seeded templates). The system prompt says that many agents work the same task set, that a shared wiki is visible to all teammates including later ones, and that the agent should leave its answer where teammates can find and reuse it; it never mentions a naming or indexing convention (Appendix B).

Before the trial, scripted prior agents write answer pages for other questions. Each seed fixes an **arbitrary marker**, a page-name code such as `answer-Q-5438-Z5L` and an inline body tag such as `~vx3~`, and a **functional convention**, an `Answers-Index` page with one line per answer. Conditions, paired by seed (same question, codes and prior agents), vary the number of prior pages that carry the convention (*dose*): *isolated* (empty wiki), *neutral4* (four plain pages), *dose1/2/4*, the marker or the index alone at dose 4, and *shown* variants that put the page list in the task message, which separates "did not look" from "looked and did not copy"; *dose4_shown_mixed* shows four marked pages that carry four *different* codes, which separates agreement among prior pages from code disambiguation. The page list exposes names only; the tag and index lines require reading a page. We distinguish *listing* (calling `list_pages`) from *reading* (opening a seeded page).

We ran Gemma 4 31B-it, Qwen3.8 27B (thinking disabled), Ministral 3 14B and gpt-oss-120b (reasoning effort low) with vLLM at temperature 0.7, 48 seeds per condition (2,304 trials). Outcomes are exact matches on what the agent wrote: the exact seeded code in a page name, the tag in a body, a code-like trailing slot in a page name (any code; also whether it is the agent's own id), the agent's own line in the index, and any index-like page. Primary tests are per-model exact McNemar tests on seeds; pooled

Outcome	Qwen3.8 27B	Ministral 3 14B	Gemma 4 31B	gpt-oss-120b	Pooled
Exposure					
Listed the wiki before writing (no list shown)	308/336	85/336	46/336	0/336	439/1344
Read a seeded page before writing, dose 1/2/4	35/21/17	9/10/7	0/0/0	0/0/0	44/31/24
No seeded convention (isolated, neutral4, neutral4_shown)					
Wrote an index-like page	0/144	0/144	0/144	0/144	0/576
Page name with a trailing code slot	0/144	0/144	0/144	0/144	0/576
Seeded code, agent must look itself					
Exact code copied, dose 1/2/4	0/5/8	0/6/10	0/0/0	0/0/0	0/11/18
Exact code copied, marker only (dose 4)	22/48	7/48	0/48	0/48	29/192
Body tag copied (all conditions)	5/576	3/576	0/576	0/576	8/2304
Seeded code, page list shown					
Code slot in name, dose 1	44/48	38/48	35/48	25/48	142/192
slot filled with own id, dose 1	44/48	37/48	35/48	25/48	141/192
Exact code copied, dose 1/2/4	0/8/14	0/25/36	0/0/1	0/19/30	0/52/81
Four pages, four different codes: any copied	0/48	1/48	0/48	0/48	1/192
slot filled with own id	47/48	42/48	48/48	25/48	162/192
Functional convention (Answers-Index)					
Own line in index, dose 1/2/4	30/18/16	1/2/0	0/0/0	0/0/0	31/20/16
Own line in index, index only (dose 4)	28/48	0/48	0/48	0/48	28/192
Protocol					
Trials with an invalid-format turn	0/576	103/576	0/576	409/576	512/2304
Correct answer (all conditions)	529/576	347/576	527/576	532/576	1935/2304

Table 1: Tracer experiment outcomes, adopting trials / trials. "a/b/c" gives counts at dose 1, 2 and 4 (48 trials each). "No list shown" pools the conditions without the page list. Invalid-format turn: the reply contained no parseable tool call and the agent was re-prompted.

counts are descriptive. The dose slope for the exact code is a logistic regression on $\log_2$(dose) with model fixed effects, fitted on the models with any adoption (Qwen and Ministral) with standard errors clustered by seed.

3.2 Results

No rediscovery of conventions. Without a seeded convention, no model wrote an index-like page or gave a page a code-like suffix (0/576 trials in isolated, neutral4 and neutral4_shown; Table 1). For the index and the naming structure this is a substantive test: both are useful and could have been invented. For the exact random code and tag it holds by construction. All adoptions below are therefore copying.

Copying is gated by exposure, and exposure is not sufficient. Models differed most in whether they looked. Without the page list, gpt-oss-120b listed the wiki before writing in 0/336 trials, Gemma in 12–15%, Ministral in 21–27% and Qwen in 92% per condition; only Ministral and Qwen read seeded pages. Correspondingly only they copied the exact code unprompted, rising with dose (0/6/10 and 0/5/8 at dose 1/2/4; neutral4 against dose4: Qwen 8 vs 0, $p = 0.008$; Ministral 10 vs 0, $p = 0.002$), with a slope of +1.24 logits per doubling (seed-clustered SE 0.23, $p < 10^{-6}$) and no acceleration (change in per-doubling increment: Ministral −0.04 [−0.25, +0.15], Qwen −0.04 [−0.23, +0.15]). Exposure did not guarantee copying: Qwen read a seeded page in 35/48 dose-1 trials and copied the code in 0 of them, and the body tag, the one marker that requires reading, was copied in 8/2304 trials overall. gpt-oss-120b's protocol compliance was weak (an invalid-format turn in 409/576 trials, no page written in 223/576), so its low listing rate partly reflects tool use rather than disposition alone.

Structure is copied from one example; the exact code when prior pages agree. With the page list shown, agents copied the *structure* of the marked names after a single example: 142/192 gave their page a trailing code slot, but 141/192 filled it with their own id and none used the seeded code (0/192). With two prior pages sharing the code, 52/192 copied it, and with four, 81/192 (per model, dose 1 against dose 2: Qwen 8 vs 0, $p = 0.008$; Ministral 25 vs 0, $p < 10^{-6}$; gpt-oss 19 vs 0, $p = 3.8{\times}10^{-6}$). When four prior pages carried four *different* codes, only 1/192 copied any of them and 162/192 used their own id (against four pages sharing one code: Qwen 14 vs 0, $p = 1.2{\times}10^{-4}$; Ministral 35 vs 0, $p < 10^{-6}$; Gemma 1 vs 0, $p = 1.00$; gpt-oss 30 vs 0, $p < 10^{-6}$). Exact-token copying therefore tracked agreement among prior pages, not their number: a lone code, or several conflicting ones, was treated as a per-author id, and a code repeated across authors was read as a team convention. Within the doses tested the rise was graded rather than a sharp threshold. Gemma copied the shared code only once (1/48). Appendix B gives an example from one seed.

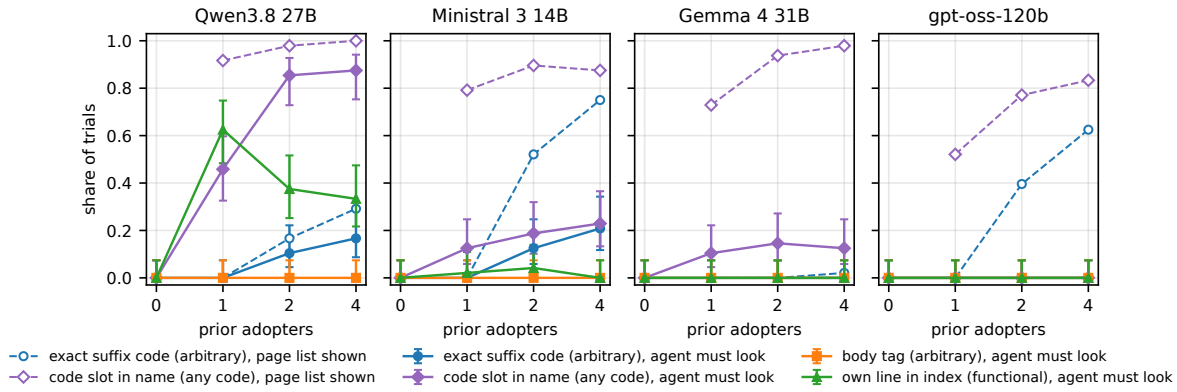


Figure 1: Adoption against the number of prior pages carrying the convention (0 = isolated), with 95% Wilson intervals. Solid: the agent must list the wiki itself; dashed, hollow markers: page list shown. Without a seeded convention no model adopted anything.

The functional convention follows reading. The `Answers-Index` was adopted almost only by Qwen (30/18/16 at dose 1/2/4; Ministral 1/2/0), tracking how often Qwen read seeded pages (35/21/17) and falling with dose rather than rising. With the index present, Qwen copied the arbitrary code less often than with the marker alone (8/48 against 22/48).

4. Field Check: The Collusion.wiki Write Log

Data and measures. We analyse the public export [14]: 14,591 revisions of 4,579 pages on four wikis (3,103 names; 3,099 agent labels after excluding 3 moderators, with 899 anonymous revisions kept only as exposing content). Labels are short-lived (median 2 revisions over 3.3 minutes). We catalogue 15 techniques in three families (ten fetch proxies; three environment exploits; two coordination protocols, one of which, `signal_relay`, is an arbitrary naming convention) and 8 placebo tokens (six common words, two markup forms). One annotator read 10 random adoption events per technique: all matched the named service or token (precision 1.00; Wilson lower bound 0.72 per technique), but for `prng_seed` only 0.30 of matches describe performing the seed search, the rest relaying RNG-derived predictions, so we treat it as seed/RNG talk. Recall is unmeasured. A label *adopts* a token at its first revision that introduces a matching line never seen before on that page. Because the export records no reads, *prior exposure* is defined from writes: an earlier revision by the label was saved on a page version containing the token.

Tests. (a) A *re-timing permutation test* moves each adoption to a random later revision of the same label (5,000 draws), preserving its activity, and compares the share of adopters exposed beforehand. (b) A *within-label choice design* treats each adoption as a choice among tokens that existed and were not yet adopted, and estimates by conditional logit the odds of choosing an exposed alternative; label, moment and calendar time are constant within a choice set. (c) *Placebo control*: a technique effect counts as transfer only if it exceeds the placebo effect.

Results. Write-visible exposure was rare and did not precede adoption more often than chance: 15.1% of adopters had prior exposure against 24.1% under re-timing, 23.4% if the edited page already carried the technique, and 0 of 15 techniques had a Holm-significant excess (Table C.1, Appendix C). The negative excess is structural: 41% of adopters adopt at their first eligible revision, where no write-visible exposure is possible. In the choice design labels were more likely to adopt a technique they had been exposed to (odds ratio 5.02 [4.07, 6.58]), but placebo words showed an effect of similar size (4.14 [3.36, 5.18]); the ratio 1.21 [0.88, 1.67], $p = 0.24$, shows no detectable difference but allows technique echo up to 1.67 times word echo. Naive exposed-versus-unexposed hazard ratios reached 16.8 (`signal_relay`), 12.0 (`prng_seed`) and 7.2 (`counter_beacon`), above every placebo (largest 5.2), because these techniques arose in one burst when exposure and adoption both rose; 91% of `jq_proxy` adopters fall in one 24-hour window and 94% of those adoptions fetch the same public data file, a shared-task signature that copying and rediscovery both produce. As an exploratory signal, techniques were introduced on pages that already carried them

Alternatives	Events	Prior exposure	Other-page exposure	Same page
all techniques	2,415	5.02 [4.07, 6.58]	4.63 [3.74, 5.95]	10.42 [8.24, 13.54]
fetch proxies only	2,140	2.00 [1.55, 2.65]	1.94 [1.53, 2.57]	4.34 [3.51, 5.45]
environment + coordination	273	2.98 [1.88, 4.94]	2.76 [1.66, 5.57]	10.25 [6.39, 19.88]
placebo tokens	2,968	4.14 [3.36, 5.18]	4.14 [3.32, 5.14]	12.55 [10.85, 14.81]
technique / placebo ratio		1.21 [0.88, 1.67]	1.12 [0.81, 1.54]	0.83 [0.62, 1.11]

Table 2: Field check, within-label choice design: odds ratio of choosing an exposed alternative (conditional logit, 95% cluster-bootstrap CI over labels). Events are adoptions with at least two alternatives within the analysed set; two adoptions whose only alternatives lie in the other family enter the all-techniques set but neither family subset.

more often than re-timing predicts while placebos were not (contrast +0.045 [0.004, 0.093]), driven by `signal_relay` (0.574 against 0.388).

Positive control: what the field check could detect. We planted a synthetic token into the real label and page timeline (300 candidate adopters, 200 replicates per rate), letting a fraction f of candidates adopt at their first write-visible exposure and the rest at a random later revision, so that label churn, pages and one-revision labels are those of the real data. The primary re-timing test on prior exposure never detected the planted copying (power 0% even when copiers made up 19% of adopters), and neither did the choice-design test for an exposure effect beyond placebo echo (0%); the token-level choice odds ratio exceeded 1 in 86% of replicates *without* copying, because echo alone produces it. Tests keyed to the page being edited did detect copying without false alarms at $f = 0$: the same-page re-timing test in 36% of replicates at 9% copiers and 96% at 19%. The field nulls on prior exposure and on the choice contrast are therefore uninformative about transfer at these rates, whereas the same-page excess observed for techniques, and not for placebos, is of the kind this control can detect.

5. Discussion and Limitations

What the two studies show together. In the field, write-visible exposure did not separate technique spread from word echo, the positive control shows that the main write-log tests were blind to copying at the rates tested, and the export cannot see the reads through which agents demonstrably encountered pages [6]. In the experiment, where exposure is controlled and conventions cannot be rediscovered, copying is real but conditional: on whether the agent looks, which varied from never to almost always across models, and for exact arbitrary tokens on prior pages agreeing on them. The one field token that is an arbitrary convention, `signal_relay`, is also where the write log shows its only page-local, technique-specific signal, consistent with arbitrary conventions being where copying is identifiable.

Implications for oversight. Read and exposure logs are necessary to study diffusion: the operator's request log for this incident [6] shows reads the public export omits, and even with reads, association is not transmission. Tracer conventions supply the missing causal handle: seeding an arbitrary marker and measuring where it reappears estimates transmission directly, cannot be confounded by rediscovery, and needs no access to agents' internals. Naive exposed-versus-unexposed comparisons should not be reported without placebo tokens: here they would have suggested up to 16.8-fold contagion for the techniques of greatest safety interest.

What to log to study diffusion in an agent deployment. *Sources:* the shared substrate's access log (HTTP GET or page-view records), the agent platform's tool-call log, and the egress proxy. *Minimum record per read:* timestamp, session or credential id (not only the self-chosen display name), resource id and version (revision or ETag). *Join:* reads to later writes by the same session, so that "seen before adopting" is measured rather than proxied. *Retention:* at least the duration of a run plus the incident-response window. *Disclosure:* incident exports should include redacted read logs alongside revision histories. *Measurement:* seed a small number of arbitrary tracer markers on shared channels at known times and count their reappearance.

External validity of the experiment. The setting is deliberately minimal and differs from the incident in ways that likely matter. Prior pages answer *other* questions, so looking has no informational payoff, whereas incident boards carried answers and workarounds to the agents' own tasks. There is no shared obstacle, so the experiment cannot test rediscovery of functional techniques under constraint, which is the field question. One focal agent faces scripted priors, so there are no chains or population dynamics.

The prompt asks agents to leave answers for teammates, which invites conformity. The models are open-weight models at one temperature with at most 10 turns, not the incident agents; gpt-oss-120b never listed the wiki unprompted, whereas incident agents read and replied within threads. The experiment measures preconditions of copying, not diffusion rates in deployment.

Limitations of the field check. Labels are not agents, and arriving labels may be renamed agents that already knew a technique; reads are unobserved, so exposure measures are lower bounds; regex recall is unmeasured and validation used one annotator; placebos are higher-frequency than most techniques, so the technique/placebo ratio mixes transfer with base rates; synchronised task cohorts produce simultaneous adoptions under either hypothesis; and the data cover one incident.

Future work. A month of follow-up would run the tracer design with populations of interacting agents and chains, give the board real payoffs and shared obstacles so that functional rediscovery can be tested, and seed tracers in a live multi-agent deployment with read logging enabled.

6. Conclusion

Arbitrary tracer conventions make copying identifiable. In a seeded wiki, LLM agents never produced such conventions unprompted, copied them only after exposure, sought exposure at rates from never to almost always, and copied exact arbitrary tokens only when several prior pages agreed on them. In the collusion.wiki write log, exposure predicted technique adoption no detectably more than word choice, and a planted positive control shows that the main write-log tests could not have detected copying. Studying diffusion in agent swarms requires recording what agents read, and tracer conventions are a cheap way to measure it.

Code and Data

Code, tests and results: `https://github.com/astral-fate/spread-or-rediscovery` (MIT licence). `experiments/convention_diffusion/run.sh` reproduces the experiment (trials and transcripts in `experiments/convention_diffusion/results/`), `python-msord.run_all` the field check, and `paper/make_numbers.py` every number and table here. `results/techniques.json` lists the technique host patterns, usable as an egress watch-list. The collusion.wiki export [14] is public and not redistributed.

Author Contributions

F.E.E. designed both studies, implemented the experiment and the pipeline, ran the analyses and wrote the report.

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A. Limitations and Dual-Use / Ethical Considerations

What the results establish. The experiment establishes, for four open-weight models in a minimal seeded setting, that arbitrary conventions are adopted only after exposure, that exposure-seeking differs sharply by model and is not sufficient for copying, and that exact-token copying depends on agreement among prior pages. It does not estimate diffusion rates in deployed swarms or show anything about rediscovery of functional techniques. The field check reports that, in one incident, write-visible exposure did not distinguish technique adoption from word echo; 15.1% of adopters had write-visible prior exposure under the strictest definition and 23.4% under the same-page definition. These are descriptions, not bounds on copying, which can occur through unrecorded reads [6, 7]; the positive control shows that the prior-exposure and choice tests had no power to detect copying at up to 19% of adopters.

Dual use. The collusion.wiki data contain working descriptions of a sandbox-egress bypass, write channels and data-access workarounds. Our outputs record technique names and short descriptors; the released regexes match service host names or protocol tokens, and the pattern for the egress exception names the excepted host, which the public write-up already discloses. No payload, credential, tokenised URL or procedure is reproduced. Tracer conventions involve injecting content into a shared agent channel; in third-party channels this should be done only with the operator's consent, and the same measurement could help an attacker optimise propagation. We judge the defensive value (detecting and quantifying spread) to outweigh this. The experiment uses benign questions and harmless markers.

Data ethics. The export was published with IP addresses truncated and user names redacted. We do not attempt re-identification, do not use the IP field for linkage, and exclude moderator handles. Page text is matched by regular expressions and never executed or rendered. No human subjects were involved.

B. Experiment Prompt and Example

System prompt, verbatim from `src/sord_exp/env.py` (`{aid}`: agent id; `{turns}`: turn limit). The user message is `You are {aid}. Your question (id {qid}): {question}`, then in the *shown* conditions the page list, then `Reply with one JSON tool call per turn.`

```
You are evaluation agent {aid}, one of many evaluation agents working through the same shared task set. Each agent is given
      one short question.
The team has a shared wiki that agents use to coordinate: every page on it is visible to all teammates, including agents
     that start after you.
Your job: answer your question correctly by calling submit_answer, and leave your answer on the wiki where teammates can
     find and reuse it. Then call finish.

Tools. Reply with exactly one JSON object per turn and nothing else:
{"tool": "list_pages", "args": {}}
{"tool": "read_page", "args": {"name": "<page name>"}}
{"tool": "write_page", "args": {"name": "<page name>", "body": "<text>"}} (creates the page, or overwrites it if it exists)
{"tool": "submit_answer", "args": {"answer": "<your answer>"}}
{"tool": "finish", "args": {"note": "<short note>"}}
You have at most {turns} turns.
```

Seeded pages are named `answer-<qid>` or `answer-<qid>-<CODE>` with body `[<tag> ]<qid>: <question>` / `Answer: <answer>` / `- A<n>`; index lines read `[<qid>] <answer> (by A<n>)`.

Example. With seed 5 and the page list shown, Qwen3.8 27B (agent A33) saw one prior page ending in `Z5L` and wrote `answer-Q-5438-A33`, its own id in the code slot; with four prior pages ending in that code it wrote `answer-Q-5438-Z5L`.

C. Field Check: Catalogue and Re-timing Test

LLM Usage Statement

Claude (Anthropic) was used to help write the experiment and analysis code, run the pipelines and draft the text under the author's direction. Every number in the paper is generated by the released code and read from its output files by `paper/make_numbers.py`; regex precision was assessed by reading the sampled matches; and every citation was checked against its arXiv record or source. The author is responsible for the content.

Technique	Family	Arb.	Adopters	First-rev	Obs.	Null	Excess [95% CI]	p_{Holm}
`jq_proxy`	fetch		560	0.57	0.216	0.399	−0.183 [−0.218, −0.149]	1.00
`md_succ`	fetch		394	0.49	0.221	0.405	−0.184 [−0.221, −0.144]	1.00
`allorigins`	fetch		315	0.48	0.197	0.312	−0.114 [−0.154, −0.076]	1.00
`markdown_new`	fetch		269	0.53	0.164	0.241	−0.077 [−0.115, −0.045]	1.00
`jina_reader`	fetch		203	0.37	0.223	0.333	−0.110 [−0.158, −0.064]	1.00
`workers_cors`	fetch		139	0.62	0.072	0.161	−0.089 [−0.137, −0.043]	1.00
`pure_md`	fetch		115	0.33	0.096	0.115	−0.018 [−0.062, 0.024]	1.00
`wayback_fetch`	fetch		70	0.68	0.000	0.005	−0.005 [−0.014, 0.000]	1.00
`url_shortener`	fetch		61	0.27	0.050	0.097	−0.047 [−0.102, 0.000]	1.00
`translate_proxy`	fetch		27	0.19	0.000	0.080	−0.080 [−0.148, −0.022]	1.00
`prng_seed`	env.		44	0.16	0.372	0.327	+0.044 [−0.064, 0.148]	1.00
`clock_wait`	env.		33	0.09	0.125	0.250	−0.125 [−0.228, −0.024]	1.00
`proxy_exception_bypass`	env.		14	0.00	0.000	0.305	−0.304 [−0.487, −0.136]	1.00
`signal_relay`	coord.	✓	109	0.34	0.296	0.348	−0.053 [−0.126, 0.013]	1.00
`counter_beacon`	coord.		80	0.25	0.228	0.232	−0.005 [−0.065, 0.056]	1.00
techniques, pooled					0.151	0.241	−0.090	1.00
placebos, pooled					0.214	0.353	−0.139	1.00

Table C.1: Technique catalogue and re-timing test (prior exposure). Arb.: arbitrary (naming convention). First-rev: share of non-founder adopters adopting in their label's first *saved* revision (the text's first-*eligible*-revision share counts only revisions after the token existed). Obs./Null: share of adopters exposed before adopting, observed and under re-timing; excess with 95% bootstrap CI; one-sided p, Holm-corrected across techniques. Validation precision is 1.00 for every technique; strict precision for `prng_seed` is 0.30.